Cover letter

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Unsent submission correspondence; prepared for author review.

Dear Editors,

Please consider the manuscript “Even-intersecting families of lattice paths and the Catalan numbers” by Deep Bhattacharjee.

The paper proves the Catalan classification of maximum even-intersecting North-East lattice-path families. Its complete written induction classifies all equality cases of the underlying weighted binary-word inequality by noncrossing partial matchings, and excludes the exceptional pair in every admissible grouping.

The classification of equality cases, extremal set/path families and Catalan enumeration fit enumerative and extremal combinatorics.

The prior public preprint is `https://hexagonmath.org/2610.00474v1`. The research manuscript/source archive is `https://doi.org/10.5281/zenodo.23608134`. Editable sources and verification records are supplied with the manuscript.

The Lean checks cover stated finite ranges, including path-family classification for n at most 3. The all-length classification is justified by the written induction and is not claimed to be fully formalised in Lean.

The manuscript and the accompanying declaration disclose the substantive use of Claude and OpenAI tools. The submitting author has declared that all parts have been checked and understood by a human. The named human authors retain responsibility for the content.

Sincerely,

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